

1. Introduction

Employee attendance and working-hour data can provide useful information about workforce availability, productivity, punctuality, overtime, and work patterns. This project analyzes employee attendance and login/logout data for the Bangalore workforce during Q1 2026.

The dataset contains 55,374 attendance records for 980 unique employees working across five major office locations in Bangalore. The available information includes department, employment type, office location, shift type, attendance status, work mode, working hours, productive hours, late arrival, early exit, and overtime.

The main goal of the project is to convert raw attendance data into useful HR and operational insights. Python was used for data inspection, cleaning, exploratory analysis, and statistical testing, while Looker Studio was used to create the final interactive dashboard. The project development process follows a structured pipeline of data inspection, data cleaning, analysis, statistical testing, and dashboard reporting.

2. Project Objectives

The main objectives of this project are:

- 1. To understand overall employee attendance and leave patterns.
- 2. To compare attendance and leave rates across departments.
- 3. To analyze workforce distribution by employment type, office location, department, and shift.
- 4. To understand how different work modes are distributed across departments.
- 5. To compare productive working hours across departments and work modes.
- 6. To identify overtime and punctuality patterns.
- 7. To present the findings through an interactive three-page HR analytics dashboard.

3. Dataset Overview

The raw dataset contains 55,374 rows and 22 columns. The project guide defines checks for data shape, column structure, duplicate records, whitespace issues, and missing values before analysis.

Important fields used in the analysis

Field	Description
employee_id	Unique employee identifier
department	Employee department
employment_type	Employment category
office_location	Employee office location
attendance_date	Attendance date
attendance_status	Present, Half Day, or On Leave
shift_type	Employee shift schedule
work_mode	Work From Office, Work From Home, Client Site, or Not Applicable
total_hours_worked	Total time worked during a shift
net_productive_hours	Productive hours after break deductions
late_arrival_mins	Minutes of late arrival
early_exit_mins	Minutes of early exit
overtime_hours	Overtime hours recorded

The project documentation also confirms that missing login and logout timestamps are linked to leave records and are therefore structurally valid.

4. Data Cleaning and Preparation

Before analysis, the dataset was checked and prepared to make the data reliable for reporting.

The main preparation steps included:

- Checking the structure and size of the dataset.
- Checking duplicate records.
- Checking missing values.
- Removing extra spaces from text fields.
- Standardizing categorical values.
- Converting date and timestamp fields into the correct format.
- Recalculating working-hour and productive-hour values.
- Making sure productive hours were not negative.
- Exporting the prepared dataset for further analysis.

The development guide specifically describes standardizing categorical columns, converting timestamps to datetime format, recalculating gross and net productive hours, and enforcing non-negative productive-hour values.

5. Key Performance Indicators

The dashboard uses a common KPI section to provide a quick summary of the workforce.

Key Indicator	Performance	Dashboard Value	Formula / Definition	Business Significance
Total Employees		980	COUNT_DISTINCT(employee_id)	Shows the total workforce covered by the analysis.
Attendance Rate		95.24%	(Present + Half Day) / Total Records	Shows the overall attendance level.
Leave Rate		4.76%	On Leave Records / Total Records	Shows the overall share of leave records.
Half Days		1,186	Count of Half Day records	Shows the number of partial-attendance records.
Avg Productivity Ratio		88.34%	AVG(productivity_ratio)	Shows the average share of working time recorded as productive time.
On-Time Arrival Rate		50.37%	On-Time Worked Records / Worked Records	Shows the proportion of worked records with on-time arrival.
Avg Productive Hours		8.12 hrs	AVG(net_productive_hours) on worked records	Shows average productive working time.
Total Overtime Hours		8,308.79 hrs	SUM(overtime_hours)	Shows total overtime recorded during Q1 2026.

The project guide also lists employee headcount, attendance rate, productive hours, productivity ratio, late arrival, early exit, overtime, WFH share, and leave count as core dashboard metrics.

6. Main Calculated Metrics

Several calculated metrics were used to make the dashboard more meaningful.

6.1 Attendance Rate

Attendance Rate = (Present + Half Day) / Total Records

This metric shows the overall attendance participation level.

6.2 Leave Rate

Leave Rate = On Leave Records / Total Records

This metric is used for comparing leave behavior across departments.

6.3 On-Time Arrival Rate

On-Time Arrival Rate = On-Time Worked Records / Total Worked Records

Only worked records are considered, so leave records are not counted as either late or on-time.

6.4 Average Productive Hours

This is the average `net_productive_hours` across worked records.

6.5 Productivity Ratio

Productivity Ratio = Net Productive Hours / Total Hours Worked

The project documentation identifies productivity ratio as a direct measure of efficiency based on productive time relative to gross working time.

6.6 Overtime Rate

Overtime Rate represents the proportion of worked records where overtime was recorded.

6.7 Average Late Arrival and Early Exit

These measures represent the average values of `late_arrival_mins` and `early_exit_mins` for worked records.

7. Analytical and Statistical Findings

The analysis included both descriptive analysis and statistical testing. Five main tests were used to study work mode, productivity, punctuality, shift patterns, and office attendance behavior.

7.1 Work Mode and Productive Hours

A Welch's independent two-sample t-test was used to compare net productive hours between Work From Office and Work From Home employees. The original analysis found a statistically significant difference between the two groups, with WFH showing higher average net productive hours.

7.2 Total Hours Worked and Productive Hours

Pearson correlation was used to study the relationship between total hours worked and net productive hours. The analysis reported a very strong positive correlation of $r = 0.9886$ with $p < 0.001$.

7.3 Work Mode and Late Arrival

A Welch's t-test was used to compare late arrival minutes between WFO and WFH records. The original analysis found a highly significant difference, with remote workers showing lower late-arrival minutes.

7.4 Shift Type and Productivity

A one-way ANOVA found significant differences in net productive hours across the five shift types, with $F = 22.45$ and $p < 0.001$. The original analysis reported Flexible shifts as having the highest average productivity and Early shifts as the lowest.

7.5 Office Location and Attendance

A Chi-Square test found no statistically significant relationship between office location and attendance status, with $p = 0.952$. This suggests that attendance behavior was broadly similar across the five office locations.

8. Looker Studio Dashboard

The final dashboard is divided into three pages, with each page focusing on a different business area.

8.1 Page 1 — Attendance & Leave

Purpose :

This page focuses on employee attendance, leave behavior, attendance trends, and punctuality.

8.1.1 Attendance Status Distribution

Chart Type: Donut Chart

Dimension: attendance_status

Metric: Record Count

This chart shows the overall distribution of Present, Half Day, and On Leave records. It is useful for quickly understanding the overall attendance composition.

In the final dashboard, Present records account for 93.1% (51,553 records) of all attendance records, On Leave for 4.76% (2,635 records), and Half Day for the remaining 2.1% (1,186 records). The composition confirms that the vast majority of the quarter was covered by full working days.

8.1.2 Monthly Attendance Composition

Chart Type: 100% Stacked Column Chart

Dimension: Month

Breakdown: attendance_status

Metric: Record Count

This chart compares the Present, Half Day, and On Leave composition across January, February, and March.

In the final dashboard the composition is almost identical in all three months, with Present remaining at roughly 93% and On Leave and Half Day making up the small remainder. This shows that attendance behavior was stable through the quarter and that no single month drove the overall attendance or leave figures.

8.1.3 Leave Rate by Department

Chart Type: Horizontal Bar Chart

Dimension: department

Metric: Leave Rate

This chart compares leave rates across departments. It helps identify departments with relatively higher or lower leave levels.

The leave rates shown in the final dashboard are:

- Product Management — 5.32%
- Data Science — 5.16%
- Finance — 5.04%
- Customer Success — 4.98%

- Sales — 4.71%
- Marketing — 4.70%
- Engineering — 4.63%
- Operations — 4.62%
- Human Resources — 4.36%
- Design — 4.11%

The distance between the highest department (Product Management) and the lowest department (Design) is only about 1.21 percentage points, so leave behavior is fairly even across the organization rather than being concentrated in one or two teams.

8.1.4 On-Time Arrival Rate by Shift Type

Chart Type: Horizontal Bar Chart

Dimension: shift_type

Metric: On-Time Arrival Rate

This chart compares punctuality across different shift schedules and helps identify shifts where on-time arrival is stronger or weaker.

The on-time arrival rates shown in the final dashboard are:

- Early (08:00–17:00) — 52.89%
- Late/US Overlap (14:00–23:00) — 49.03%
- Mid (11:00–20:00) — 47.98%
- General (09:30–18:30) — 47.87%
- Flexible (10:30–19:30) — 44.94%

Early shift records show the strongest punctuality and Flexible shift records the weakest. However, every shift stays below 53%, which matches the overall on-time arrival rate of 50.37% and indicates that punctuality is an organization-wide issue rather than a single-shift issue.

8.1.5 Weekly Attendance Rate Trend

Chart Type: Time Series Chart

Dimension: attendance_date by ISO Year Week

Metric: Attendance Rate

This chart shows how the attendance rate changes from week to week during Q1 2026. It helps identify periods where attendance improves, declines, or remains stable.

The dashboard shows the weekly attendance rate staying inside a narrow band of 94.3% to 96.1% across the whole quarter. The weakest weeks are those beginning 12 January 2026 (94.3%) and 9 February 2026 (94.4%), and the strongest is the week beginning 9 March 2026 (96.1%). The quarter closes at 95.5% in the week beginning 30 March 2026, so there is no sustained decline in attendance at any point in Q1 2026.

8.2 Page 2 — Workforce & Work Mode

Purpose :

This page focuses on workforce structure and employee distribution across employment types, work modes, office locations, departments, and shifts.

8.2.1 Employee Distribution by Employment Type

Chart Type: Donut Chart

Dimension: employment_type

Metric: COUNT_DISTINCT(employee_id)

This chart shows how the workforce is distributed across Full-Time, Contract, Intern, and Part-Time employment types.

The final dashboard shows Full-Time employees at 82.6% (809 employees), Contract at 8.3% (81 employees), Intern at 5.8% (57 employees), and Part-Time at approximately 3.4% (33 employees) of the 980 employees. The workforce is therefore strongly permanent, and non-full-time categories together account for less than one fifth of headcount.

8.2.2 Work Mode Distribution by Department

Chart Type: 100% Stacked Vertical Bar Chart

Dimension: department

Breakdown: work_mode

Metric: Record Count

Not Applicable records are excluded. The chart compares Work From Office, Work From Home, and Client Site patterns across departments.

In the final dashboard, Work From Office is the leading mode in every department, at roughly 78% to 80% of records for most teams. Sales and Customer Success are the two exceptions: their Work From Office share falls to around 68%, and they are the only departments with a visible Client Site share, which is consistent with their client-facing responsibilities. Work From Home accounts for most of the remaining records in every department.

8.2.3 Employees by Office Location

Chart Type: Horizontal Bar Chart

Dimension: office_location

Metric: COUNT_DISTINCT(employee_id)

This chart compares the number of employees across the five Bangalore office locations.

The employee counts shown in the final dashboard are:

- Koramangala HQ — 317
- Whitefield Tech Park — 219
- HSR Layout Hub — 172
- Indiranagar Annexe — 152
- Electronic City Campus — 120

Koramangala HQ holds close to one third of the Bangalore workforce, while Electronic City Campus is the smallest site. The five locations together account for all 980 employees.

8.2.4 Employees by Shift Type

Chart Type: Vertical Bar Chart

Dimension: shift_type

Metric: COUNT_DISTINCT(employee_id)

This chart shows how employees are distributed across the available shift schedules.

The shift distribution shown in the final dashboard is:

- General (09:30–18:30) — 422
- Flexible (10:30–19:30) — 199
- Mid (11:00–20:00) — 142
- Late/US Overlap (14:00–23:00) — 132
- Early (08:00–17:00) — 85

The General shift covers about 43% of employees, while the Early shift is the smallest group at 85 employees. This is useful context when reading the shift-level punctuality and productivity results, because the Early shift results are based on the smallest population.

8.2.5 Employee Headcount by Department

Chart Type: Treemap

Dimension: department

Metric: COUNT_DISTINCT(employee_id)

This chart gives a compact view of department size and highlights larger and smaller departments.

The department headcounts shown in the final dashboard are:

- Engineering — 292
- Sales — 146
- Customer Success — 114
- Data Science — 76
- Operations — 74
- Marketing — 67
- Human Resources — 61
- Product Management — 59
- Finance — 56
- Design — 35

Engineering alone accounts for 292 of the 980 employees, and Engineering, Sales, and Customer Success together make up more than half of the workforce. Design is the smallest department at 35 employees, so its department-level rates are based on a much smaller group and should be read with that in mind.

8.3. Page 3 — Productivity & Punctuality

Purpose :

This page focuses on productive working time, overtime, work-mode productivity, and shift-level punctuality.

8.3.1 Average Productive Hours by Department

Chart Type: Horizontal Bar Chart

Dimension: department

Metric: AVG(net_productive_hours)

Filter: total_hours_worked > 0

This chart compares the average productive working time across departments. Using the average makes the comparison fairer than comparing total productive hours.

The average productive hours shown in the final dashboard are:

- Engineering — 8.32 hrs
- Sales — 8.26 hrs
- Product Management — 8.12 hrs
- Design — 8.06 hrs
- Customer Success — 8.00 hrs
- Marketing — 8.00 hrs
- Human Resources — 7.98 hrs
- Data Science — 7.90 hrs
- Finance — 7.90 hrs
- Operations — 7.81 hrs

The full range is only about half an hour, from 7.81 hours in Operations to 8.32 hours in Engineering, so productive working time is broadly similar across departments and sits close to the overall average of 8.12 hours.

8.3.2 Overtime Rate by Department

Chart Type: Horizontal Bar Chart

Dimension: department

Metric: Overtime Rate

The metric measures the share of worked records where overtime was recorded. The chart helps identify departments where overtime occurs more frequently.

The overtime rates shown in the final dashboard are:

- Engineering — 31.7%
- Sales — 28.3%
- Product Management — 25.8%
- Data Science — 23.2%
- Customer Success — 22.0%
- Marketing — 19.9%
- Design — 19.0%
- Finance — 17.6%
- Human Resources — 16.9%
- Operations — 16.6%

Overtime varies far more between departments than productive hours do. Engineering records overtime on 31.7% of worked records, close to double the rate of Operations at 16.6%. The three departments with the highest average productive hours (Engineering, Sales, and Product Management) are also the three departments with the highest overtime rates, which suggests that their higher productive time is partly driven by longer working days rather than by efficiency alone.

8.3.3 Average Productive Hours by Work Mode

Chart Type: Vertical Bar Chart

Dimension: work_mode

Metric: AVG(net_productive_hours)

Not Applicable and non-worked records are excluded.

The current dashboard shows:

- Client Site — 8.35 hrs
- Work From Home — 8.16 hrs
- Work From Office — 8.09 hrs

This chart shows that Client Site records have the highest average productive hours among the three valid work modes in the final dashboard.

8.3.4 Average Late Arrival & Early Exit by Shift

Chart Type: Grouped Vertical Bar Chart

Dimension: shift_type

Metrics:

- AVG(late_arrival_mins)
- AVG(early_exit_mins)

Filter: total_hours_worked > 0

This chart compares late arrival and early exit behavior across shifts. Both measures use the same minutes scale so that the two metrics can be compared directly.

The averages shown in the final dashboard are:

- Flexible (10:30–19:30) — 11.23 mins late arrival, 18.25 mins early exit
- Mid (11:00–20:00) — 10.51 mins late arrival, 23.32 mins early exit
- Late/US Overlap (14:00–23:00) — 10.20 mins late arrival, 20.28 mins early exit
- General (09:30–18:30) — 10.13 mins late arrival, 19.83 mins early exit
- Early (08:00–17:00) — 9.39 mins late arrival, 31.58 mins early exit
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Average late arrival is close to 10 minutes in every shift, so late-arrival behavior differs only slightly between shifts. Early exit varies much more. The Early shift averages 31.58 minutes of early exit, well above every other shift, even though it has the lowest late arrival and the highest on-time arrival rate. This pattern is consistent with the ANOVA result in Section 7.4, which reported the Early shift as having the lowest average productive hours.

8.3.5 Productivity Ratio by Department

Chart Type: Treemap

Dimension: department

Metric: AVG(productivity_ratio)

Filter: total_hours_worked > 0

This chart compares productivity efficiency across departments. A higher productivity ratio means a larger share of recorded working time is classified as productive time.

The productivity ratios shown in the final dashboard are:

- Sales — 88.8%
- Customer Success — 88.5%
- Design — 88.5%
- Engineering — 88.5%
- Product Management — 88.3%
- Human Resources — 88.2%
- Marketing — 88.1%
- Finance — 87.9%
- Operations — 87.8%
- Data Science — 87.7%

All ten departments fall within roughly one percentage point of each other and of the overall average of 88.34%. Efficiency is therefore consistent across the organization, and the department differences seen in productive hours come mainly from how long people work rather than from how productively they use that time.

9. Key Dashboard Insights

The final dashboard provides several useful observations:

1. The overall attendance rate is high at 95.24%, with Present records making up 93.1% of all attendance records.
2. The overall leave rate is 4.76%, and 1,186 half-day records were recorded during the quarter.
3. Department-level leave rates range from 4.11% in Design to 5.32% in Product Management, a spread of only about 1.21 percentage points.
4. On-time arrival is weak across the whole organization at 50.37%, ranging from 44.94% on the Flexible shift to 52.89% on the Early shift.
5. Weekly attendance stays within a narrow 94.3% to 96.1% band across Q1 2026, with the strongest week beginning 9 March 2026.
6. The workforce of 980 employees is concentrated in Engineering (292 employees) and at Koramangala HQ (317 employees), and 82.6% of employees are Full-Time.
7. Average productive hours by department range narrowly from 7.81 hrs in Operations to 8.32 hrs in Engineering.
8. Overtime rates vary far more widely than productive hours, from 16.6% in Operations to 31.7% in Engineering, with 8,308.79 overtime hours recorded in total.
9. Client Site shows the highest average productive hours at 8.35 hrs, ahead of Work From Home at 8.16 hrs and Work From Office at 8.09 hrs.

10. Average late arrival is close to 10 minutes on every shift, while early exit ranges from 18.25 minutes on the Flexible shift to 31.58 minutes on the Early shift.
11. The productivity ratio is stable at 87.7% to 88.8% across all departments, averaging 88.34%.

10. Business Recommendations

10.1 Review Shift Allocation

Departments with lower productivity or weaker punctuality can be reviewed for better shift allocation. The statistical analysis found significant productivity differences across shift types, and the dashboard supports this: the Early shift has the best on-time arrival rate (52.89%) but by far the highest average early exit (31.58 minutes), while the Flexible shift has the weakest on-time arrival rate (44.94%). Reviewing start and end expectations for these two shifts is likely to give the largest improvement.

10.2 Monitor Punctuality

Punctuality should remain an important HR monitoring area. The overall on-time arrival rate is only 50.37%, and no shift exceeds 52.89%, so the issue is organization-wide rather than limited to one group. Average late arrival is close to 10 minutes on every shift, which suggests that formal flexible arrival windows may be a more practical response than individual follow-up.

10.3 Monitor Overtime

Departments with higher overtime rates should be reviewed for workload, staffing, and scheduling issues. Overtime is heavily concentrated: Engineering (31.7%), Sales (28.3%), and Product Management (25.8%) record overtime at close to double the rate of Operations (16.6%) and Human Resources (16.9%). Since productivity ratios are almost identical across departments, the higher overtime in these teams points to workload and staffing levels rather than to working style.

10.4 Evaluate Work-Mode Policies

The statistical analysis found a significant difference in productive hours between WFO and WFH records, and the dashboard shows Client Site highest at 8.35 hrs, followed by Work From Home at 8.16 hrs and Work From Office at 8.09 hrs. The differences are small in absolute terms, so the results support continuing flexible and hybrid work rather than treating any single mode as clearly better.

10.5 Maintain Consistent Office-Level Policies

Since the statistical analysis did not find a significant relationship between office location and attendance status, and all 980 employees are spread across five sites ranging from 317 employees at Koramangala HQ to 120 at Electronic City Campus, major attendance policies can continue to be managed consistently across locations.

11. Limitations

The analysis has a few limitations:

- The dataset covers only Q1 2026.
- The data represents attendance and working records rather than complete measures of employee performance.
- Productivity is based on recorded working and break times.
- Statistical significance does not automatically prove a direct cause-and-effect relationship.
- Some dashboard metrics use worked-day filters to avoid treating leave records as working or punctual records.

12. Conclusion

This project converts raw employee attendance and login/logout data into a practical HR analytics dashboard. The analysis combines data preparation, descriptive analysis, calculated metrics, statistical testing, and interactive visualization.

The final three-page dashboard provides a clear view of attendance and leave, workforce and work mode, and productivity and punctuality. It allows HR and management teams to compare departments, shifts, work modes, office locations, and employee groups using consistent measures.

Overall, the project demonstrates how structured data analysis and dashboard visualization can turn attendance records into useful information for workforce monitoring, productivity analysis, punctuality management, and HR decision-making.